Remote Center of Motion (RCM)

A fixed point in space, where a linkage of a mechanism pivots around. This point may or may not lie on the mechanism. This is an extended concept of Instantaneous center of motion, but the IC stays fixed in space.

This motion is used in Minimally Invasive Surgery (MIS), manufacturing etc.

Generally, there are two approaches to obtain RCM output.

1. Serial mechanism with high DOFs and ensure that the end effector mechanism makes the item pivot around a fixed point. But this requires complex control strategies and can cause problems if those fail.
2. A mechanically constrained mechanism whose RCM output is mechanically guaranteed. The Da Vinci robot is a typical example. Parallelograms and pantographs have been implemented in multiple designs.

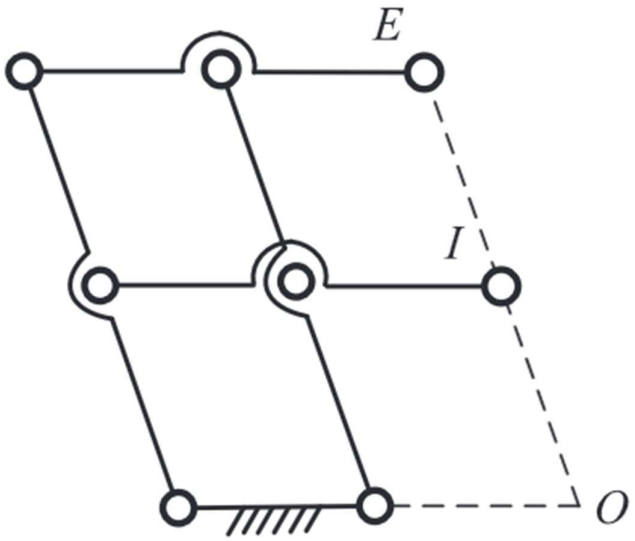


Fig 1. Structure of a parallelogram mechanism

Basic structure of a parallelogram mechanism is shown in Fig 1. It is obvious that the virtual link denoted by EI always passes through point O. Hence this link gives RCM output

Structural Design

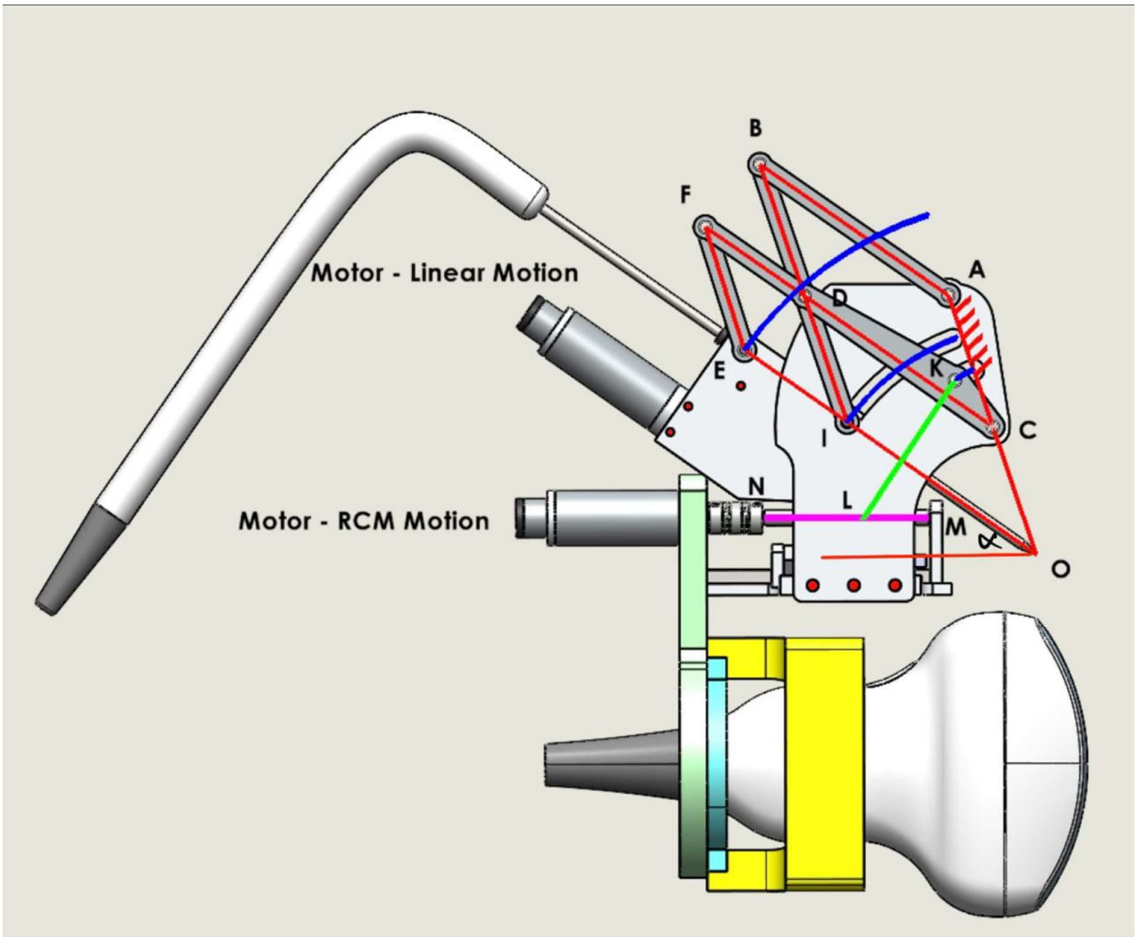


Fig 2. Kinematic linkages of the proposed design.

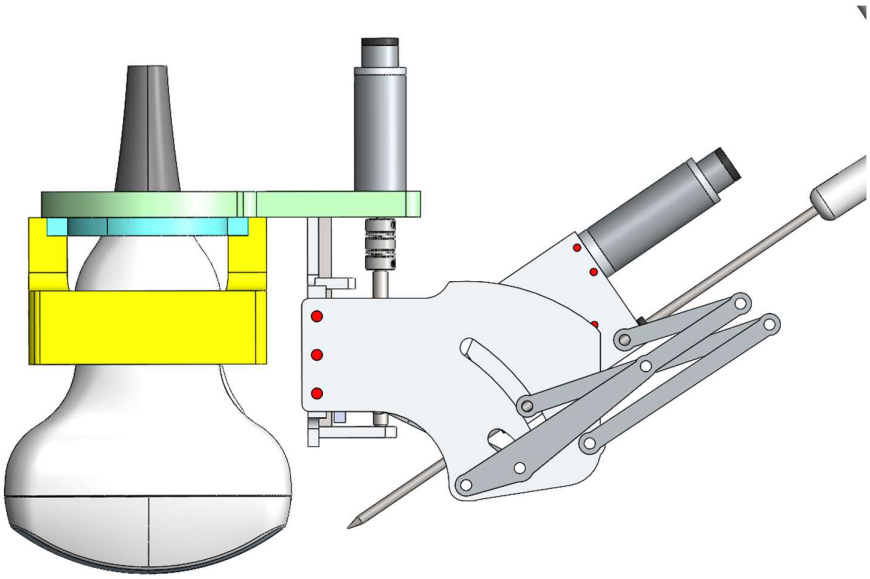


Fig 3. Linkages at the opposite extreme position

The design proposed uses a double parallelogram and the needle is passed through the virtual link EI, and the RCM lies on O. The link AC is fixed. The assembly to control the linear motion is moved by the attachment at E and slot to move the joint I.This allows the needle to pivot around RCM point O, while the linear motion can be controlled by the motor. The parallelogram is actuated by link KL connected to link KCF. The link KL is attached to a thread linear actuator assembly run by a motor, which moves L along the axis MN. As L moves towards M from the position in Fig 2., the assembly attains the form in Fig 3.

Determination of angle of needle with respect to linear position on lead screw

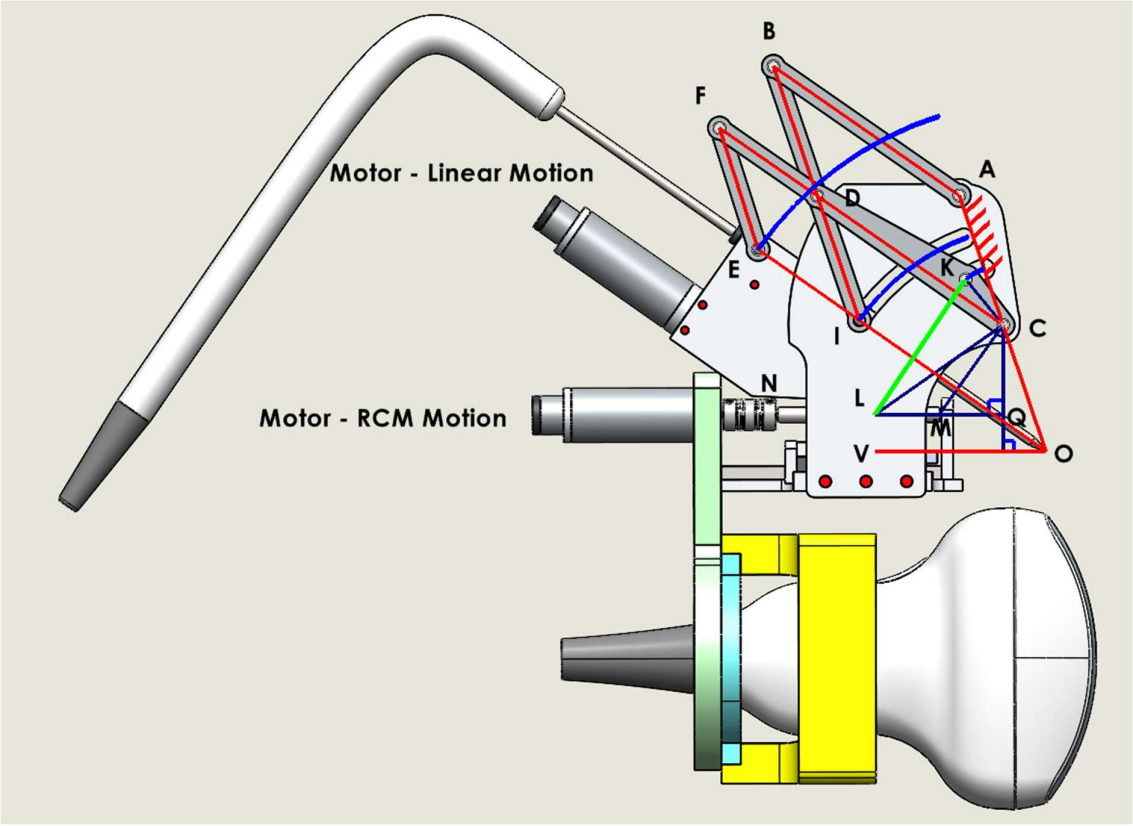


Fig 4. Linkage diagram for derivation

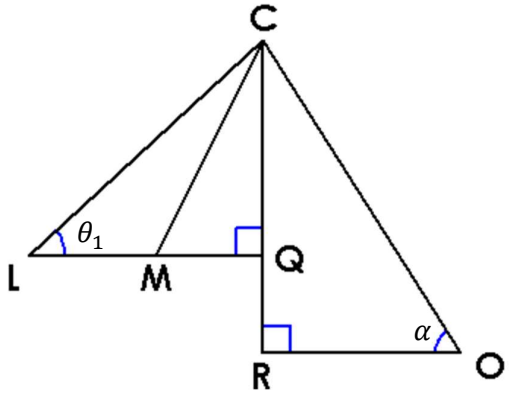
Defining of constant and variable parameters of the mechanism

The position of L can be varied with the rotation of the motor of lead screw, which will change the angle of the needle (OI) from the vertical.

Input length: - l_4, Output angle: - α

Mechanism Constraints (Does not change in operation)

AB	l_1
CK	l_2
KL	l_3
CQ	l_5
MQ	l_6
$\angle KCD$	β



Consider OR to be the vertical w.r.t the mechanism.

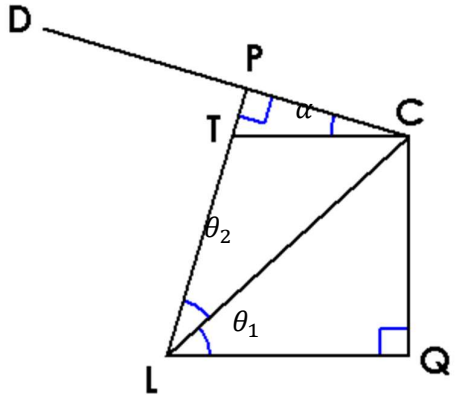
From $\Delta CRO \; and \; \Delta CQL$

$CL \; cos\theta_1 = ML + MQ$

$\theta_1 = tan^{-1}\left(\frac{CQ}{ML+MQ}\right)$ ***Equation 1.***

$CL = \frac{ML+MQ}{cos\,\theta_1}$ ***Equation 2.***

Fig 5(a). Derived $\Delta CRO \; and \; \Delta CQL$



From $\Delta PCT, \Delta TCL \; and \; \Delta CQL$

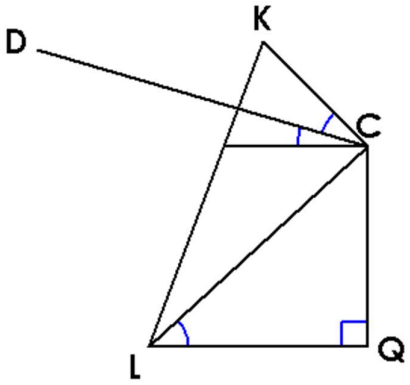
$\angle PCL = 90^0 - \theta_2$

$\angle QCL = 90^0 - \theta_1$

$(90^0 - \theta_2) - \alpha + (90^0 - \theta_1) = 90^0$

$\theta_1 + \theta_2 + \alpha = 90^0$ ***Equation 3.***

Fig 5(b). Derived $\Delta PCT, \Delta TCL \; and \; \Delta CQL$



From ΔKCL

Let $\angle KCL = \theta_3$

$\theta_3 = \beta + (90^0 - \theta_2)$ ***Equation 4.***

$KL^2 = KC^2 + CL^2 - 2.KC.KL.cos\theta_3$

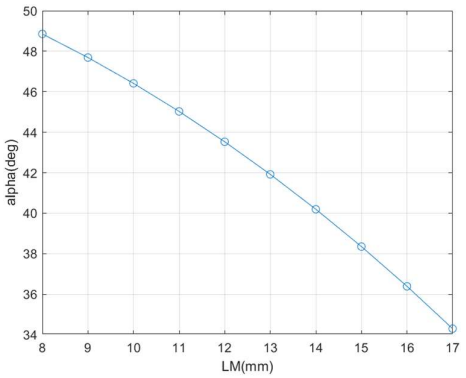
$\theta_3 = cos^{-1}\left(\frac{KC^2 + CL^2 - KL^2}{2.KC.KL}\right)$ ***Equation 5.***

Fig 5(c). Derived ΔKCL

Calculating α from LM

1. Find $\boldsymbol{\theta_1}$ using **LM** as the input using Eqn 1.
2. Find **CL** using Eqn 2.
3. Find $\boldsymbol{\theta_3}$ using Eqn 5.
4. Find $\boldsymbol{\theta_2}$ using Eqn 4.
5. Find $\boldsymbol{\alpha}$ using Eqn 3.

Relation Diagrams



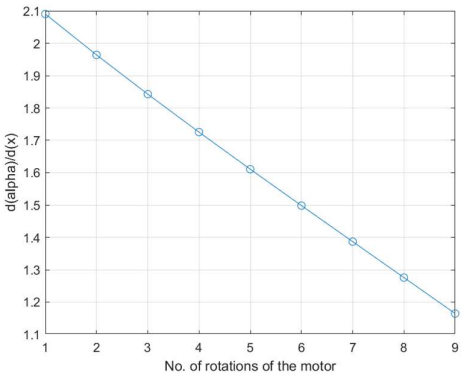


Fig 6(a). Graph: Inclination vs LM 6(b). Graph: $\frac{d\alpha}{dLM}$ vs No. of rotations of the motor (Pitch = 1mm)